Example 4.60 How many diagonals are there in a polygon with n sides?

Solution:
A polygon of n sides has n vertices. By joining any two vertices of a polygon, we obtain either a side or a diagonal of the polygon. Number of line segments obtained by joining the vertices of a n sided polygon taken two at a time is ${}^nC_2 = \frac{n(n-1)}{2}$. Out of these lines, there are n sides of polygon. Therefore, number of diagonals of the polygon is

$$\frac{n(n-1)}{2} - n = \frac{n(n-3)}{2}.$$

In particular for a pentagon and heptagon (Septagon), number of diagonals respectively are $\frac{5(5-3)}{2} = 5$ and $\frac{7(7-3)}{2} = 14$.



Exercise - 4.3

1. If ${}^nC_{12} = {}^n C_9$ find ${}^{21}C_n$.
2. If ${}^{15}C_{2r-1} = {}^{15} C_{2r+4}$, find r.
3. If ${}^nP_r = 720$, and ${}^nC_r = 120$, find n, r.
4. Prove that ${}^{15}C_3 + 2 \times {}^{15} C_4 + {}^{15} C_5 = {}^{17} C_5$.
5. Prove that ${}^{35}C_5 + \sum_{r=0}^{4} {}^{(39-r)}C_4 = {}^{40} C_5$.
6. If ${}^{(n+1)}C_8 : {}^{(n-3)} P_4 = 57 : 16$, find the value of n.
7. Prove that ${}^{2n}C_n = \frac{2^n \times 1 \times 3 \times \cdots (2n-1)}{n!}$.
8. Prove that if $1 \leq r \leq n$ then $n \times {}^{(n-1)} C_{r-1} = (n-r+1){}^nC_{r-1}$.
9. (i) A Kabaddi coach has 14 players ready to play. How many different teams of 7 players could the coach put on the court?
   (ii) There are 15 persons in a party and if each 2 of them shakes hands with each other, how many handshakes happen in the party?
   (iii) How many chords can be drawn through 20 points on a circle?
   (iv) In a parking lot one hundred , one year old cars, are parked. Out of them five are to be chosen at random for to check its pollution devices. How many different set of five cars can be chosen?
   (v) How many ways can a team of 3 boys,2 girls and 1 transgender be selected from 5 boys, 4 girls and 2 transgenders?
10. Find the total number of subsets of a set with

    [Hint: ${}^nC_0 + {}^n C_1 + {}^n C_2 + \cdots + {}^n C_n = 2^n$]
    (i) 4 elements (ii) 5 elements (iii) n elements.
11. A trust has 25 members.
    (i) How many ways 3 officers can be selected?
    (ii) In how many ways can a President, Vice President and a Secretary be selected?